Lab 6 - 3D Scan and Print a Real Object

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 01: Overview of Additive Manufacturing and 3D Scanning TSC objective: A4 - evaluate reverse engineering of a component through 3D scanning and subsequent AM

Goal

Scan a real object with a mobile 3D-scanning app, export the mesh as an STL file, repair it if needed, then slice and 3D print the reproduction.

What you'll build

A 3D-printed reproduction of a real object captured by photogrammetry on your phone.

Tools: Polycam / Kiri Engine / Scandy Pro (mobile), Fusion 360, Ultimaker Cura

Steps

1. Install a 3D scanning app on your phone - Polycam or Kiri Engine (Android/iOS), or Scandy Pro / Heges (iOS LiDAR).

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App Store / Google Play > Polycam or Kiri Engine
```

2. Pick a matte, textured object about the size of a mug and place it on a contrasting background with even lighting.

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Object: non-shiny, non-transparent, stable
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3. Capture 30-60 photos in two orbits around the object - one at eye level, one from above - keeping it centred.

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App > Photo mode > capture 2 full orbits
```

4. Let the app reconstruct the mesh, then crop away the background and floor plane.

```
App > Process > Crop / Remove floor
```

5. Export the model as an STL (or OBJ, then convert via Fusion 360's Mesh workspace).

```
App > Export > STL
```

6. Repair holes or stray triangles: Fusion 360 > Mesh > Repair, or import directly into Cura if clean.

```
Fusion 360 > Mesh > Repair > Export STL
```

7. Scale the model in Cura to a 20-minute print size (~40 mm tall), slice with supports if needed, and print.

```
Cura > Scale 40 mm > Slice > print
```

Evidence to capture

- Screenshot of the cleaned scan with the floor/background removed.
- Mesh-repair evidence showing a watertight model that loads cleanly in Cura.

- Photo comparing the source object with the recognisable scaled print.

Test it

The exported STL loads into Cura as a watertight mesh, and the printed part is a recognisable scaled reproduction of the scanned object.